MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 07 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

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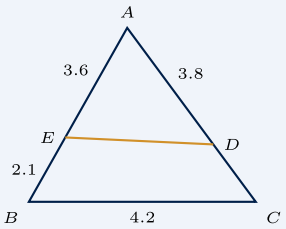
Q1. In the given figure, if $\angle ADE = \angle B$, with $AD = 3.8$ cm, $AE = 3.6$ cm, $BE = 2.1$ cm, and $BC = 4.2$ cm, then DE is equal to: [1]

(a) 2.8 cm (b) 2.1 cm (c) 3.2 cm (d) 1.9 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 2.8 cm

Given: Since segment lengths $AE = 3.6$ cm and $BE = 2.1$ cm are specified, point E lies on side AB (so $AB = AE + BE = 3.6 + 2.1 = 5.7$ cm) and point D lies on side AC, with $\angle ADE = \angle B$, $AD = 3.8$ cm, and $BC = 4.2$ cm. (*Note:* The printed thumbnail swaps labels D, E; the consistent configuration is shown on the right.)



In $\triangle ADE$ and $\triangle ABC$: $\angle A = \angle A$ (common) and $\angle ADE = \angle B$ (given), so $\triangle ADE \sim \triangle ABC$ (AA similarity; $A \leftrightarrow A$, $D \leftrightarrow B$, $E \leftrightarrow C$):

Formula / Theorem Used: $\dfrac{AD}{AB} = \dfrac{AE}{AC} = \dfrac{DE}{BC}$

$$\frac{3.8}{5.7} = \frac{DE}{4.2} \implies \frac{2}{3} = \frac{DE}{4.2} \implies DE = \frac{2 \times 4.2}{3} = \boxed{2.8 \text{ cm}}$$

Q2. In $\triangle ABC$ and $\triangle DEF$, $\angle B = \angle E$, $\angle F = \angle C$, and $AB = 3DE$. Then the two triangles are: [1]

(a) Congruent but not similar (b) Similar but not congruent (c) Neither congruent nor similar (d) Congruent and similar

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) Similar but not congruent

We know the AA Similarity Criterion and congruence condition:

Formula / Theorem Used: Two triangles are similar if two pairs of corresponding angles are equal; congruence additionally requires scale factor 1.

Since $\angle B = \angle E$ and $\angle C = \angle F$, by **AA similarity**, $\triangle ABC \sim \triangle DEF$. However, $AB = 3DE \neq DE$, so corresponding sides are unequal ($k = 3 \neq 1$). Hence, the triangles are **similar but not congruent**.

Q3. If D, E, F are respectively the midpoints of sides BC, CA, AB of $\triangle ABC$, then $\triangle DEF$ is similar to: [1]

(a) $\triangle ABC$ (b) Only $\triangle BDF$ (c) Only $\triangle CDE$ (d) None of these

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) $\triangle ABC$

Given: D, E, F are midpoints of BC, CA, AB respectively.

We know the Midpoint Theorem and SSS Similarity Criterion:

Formula / Theorem Used: $DE = \frac{1}{2}AB, \quad EF = \frac{1}{2}BC, \quad FD = \frac{1}{2}CA \implies \frac{DE}{AB} = \frac{EF}{BC} = \frac{FD}{CA} = \frac{1}{2}$

By SSS similarity, $\triangle DEF \sim \triangle ABC$. (Note that $\triangle DEF$ is congruent to all three corner triangles $\triangle AFE, \triangle BDF, \triangle CED$, so options containing "Only" are incorrect.) Thus $\triangle DEF \sim$ **$\triangle$ABC**.

Q4. In $\triangle ABC$, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 4x-3$, $DB = 3x-1$, $AE = 8x - 7$, and $EC = 5x - 3$, the value of x is: [1]

(a) 1 (b) 2 (c) $-\frac{1}{2}$ (d) 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 1

We know the Basic Proportionality Theorem ($DE \parallel BC \implies \frac{AD}{DB} = \frac{AE}{EC}$):

Formula / Theorem Used: $\frac{4x-3}{3x-1} = \frac{8x-7}{5x-3}$

Cross-multiplying:

$$(4x-3)(5x-3) = (8x-7)(3x-1) \implies 20x^2 - 27x + 9 = 24x^2 - 29x + 7$$

$$4x^2 - 2x - 2 = 0 \implies 2x^2 - x - 1 = 0 \implies (x-1)(2x+1) = 0 \implies x = 1 \text{ or } x = -\frac{1}{2}$$

Checking positivity: at $x = -\frac{1}{2}$, $AD = 4(-0.5) - 3 = -5 < 0$ (invalid). At $x = 1$, $AD = 1$, $DB = 2$, $AE = 1$, $EC = 2$ (all positive). Hence $x = \mathbf{1}$.

Q5. A scale model of a cargo ship is made to a scale of 1 : 200. If the length of the model is 4 m, the actual length of the ship is: [1]

(a) 800 m (b) 50 m (c) 80 m (d) 204 m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) 800 m

We know that the ratio of model length to actual length equals the scale factor:

Formula / Theorem Used: $\text{Scale Factor} = \dfrac{\text{Model Length}}{\text{Actual Length}} = \dfrac{1}{200}$

Let actual length be L m:

$$\frac{4}{L} = \frac{1}{200} \implies L = 4 \times 200 = \boxed{\mathbf{800}\text{ m}}$$

Q6. **Assertion (A):** Diagonals of a trapezium divide each other proportionally.
Reason (R): The triangles formed by the diagonals and parallel sides of a trapezium are similar by AA criterion. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (a) Both (A) and (R) are true and (R) is the correct explanation of (A).

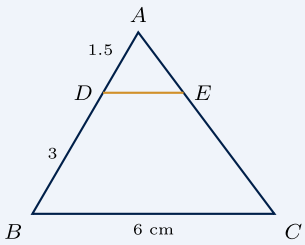
We know the Trapezium Diagonal Similarity Property ($AB \parallel DC$ with diagonals AC, BD meeting at O):

Formula / Theorem Used: $\triangle AOB \sim \triangle COD$ (by AA similarity) $\implies \dfrac{OA}{OC} = \dfrac{OB}{OD}$

In $\triangle AOB$ and $\triangle COD$, $\angle OAB = \angle OCD$ (alternate interior angles) and $\angle AOB = \angle COD$ (vertically opposite angles), so $\triangle AOB \sim \triangle COD$ (AA). Hence diagonals divide each other proportionally. Thus, both (A) and (R) are true, and (R) correctly explains (A).

Q7. In $\triangle ABC$, D and E are points on sides AB and AC such that $DE \parallel BC$. If $AD = 1.5$ cm, $DB = 3$ cm, and $BC = 6$ cm, find the length of line segment DE. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Given: In $\triangle ABC$, $DE \parallel BC$, $AD = 1.5$ cm, $DB = 3$ cm, $BC = 6$ cm. So $AB = AD + DB = 1.5 + 3 = 4.5$ cm.
We know the AA Similarity Criterion:

Formula / Theorem Used: If $DE \parallel BC$ in $\triangle ABC$, then
$$\triangle ADE \sim \triangle ABC \implies \frac{AD}{AB} = \frac{DE}{BC}.$$

Step 1: Establishing $\triangle ADE \sim \triangle ABC$ [1 Mark]
In $\triangle ADE$ and $\triangle ABC$: $\angle A = \angle A$ (common) and $\angle ADE = \angle ABC$ (corresponding angles as $DE \parallel BC$). By AA similarity, $\triangle ADE \sim \triangle ABC$.

Step 2: Computing DE [1 Mark]

$$\frac{AD}{AB} = \frac{DE}{BC} \implies \frac{1.5}{4.5} = \frac{DE}{6} \implies \frac{1}{3} = \frac{DE}{6}$$
$$DE = \frac{6}{3} = \boxed{\mathbf{2}\text{ cm}}$$

Q8. If $\triangle ABC \sim \triangle PQR$ with $\angle A = 48°$ and $\angle B = 72°$, determine the exact degree measure of $\angle R$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $\triangle ABC \sim \triangle PQR$, $\angle A = 48°$, and $\angle B = 72°$.
We know the Angle Sum Property of a triangle and the property of similar triangles:

Formula / Theorem Used: $\angle A + \angle B + \angle C = 180°$ and $\triangle ABC \sim \triangle PQR \implies \angle A = \angle P,\ \angle B = \angle Q,\ \angle C = \angle R$

Step 1: Finding $\angle C$ in $\triangle ABC$ [1 Mark]

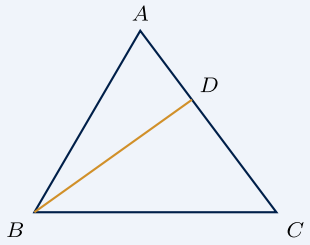
$$48° + 72° + \angle C = 180° \implies 120° + \angle C = 180° \implies \angle C = 180° - 120° = 60°$$

Step 2: Using Corresponding Angles of Similar Triangles [1 Mark]
Since $\triangle ABC \sim \triangle PQR$ with vertex correspondence $C \leftrightarrow R$:

$$\angle R = \angle C = \boxed{\mathbf{60°}}$$

Q9. In the given figure, D is a point on side AC of $\triangle ABC$ such that $\angle ABD = \angle C$. If $AB = 8$ cm and $AC = 16$ cm, calculate the length of segment AD. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: Point D on side AC of $\triangle ABC$ with $\angle ABD = \angle ACB$, $AB = 8$ cm, and $AC = 16$ cm.
We know the AA Similarity Criterion:

Formula / Theorem Used: If two pairs of angles of two triangles are equal, the triangles are similar and corresponding sides are proportional.

Step 1: Proving $\triangle ABD \sim \triangle ACB$ [1.5 Marks]
In $\triangle ABD$ and $\triangle ACB$:

1. $\angle BAD = \angle CAB$ (Common angle $\angle A$)
2. $\angle ABD = \angle ACB$ (Given)

By **AA similarity criterion**, $\triangle ABD \sim \triangle ACB$ ($A \leftrightarrow A$, $B \leftrightarrow C$, $D \leftrightarrow B$).

Step 2: Finding AD Using Proportional Sides [1.5 Marks]

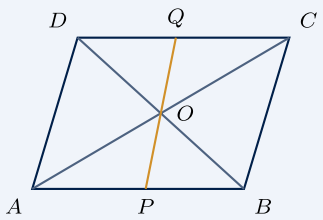
$$\frac{AB}{AC} = \frac{AD}{AB} \implies AB^2 = AD \times AC$$

Substituting $AB = 8$ cm and $AC = 16$ cm:

$$8^2 = AD \times 16 \implies 64 = 16 \cdot AD \implies \boxed{AD = \frac{64}{16} = \mathbf{4}\text{ cm}}$$

Q10. The diagonals of a parallelogram $ABCD$ intersect at O. A line through O intersects side AB at P and side CD at Q. Prove that line segment $OP = OQ$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: Parallelogram $ABCD$ with diagonals AC and BD intersecting at O; line PQ through O meets AB at P and CD at Q.
To Prove: $OP = OQ$.

We know the properties of a parallelogram and similarity/congruence criteria:

Formula / Theorem Used: In a parallelogram, opposite sides are parallel ($AB \parallel DC$) and diagonals bisect each other ($OA = OC$).

Step 1: Establishing $\triangle OAP \sim \triangle OCQ$ (Scale Factor 1) [2 Marks]

In $\triangle OAP$ and $\triangle OCQ$:

1. $\angle OAP = \angle OCQ$ (Alternate interior angles since $AB \parallel DC$, transversal AC)
2. $\angle AOP = \angle COQ$ (Vertically opposite angles)
3. $OA = OC$ (Diagonals of parallelogram $ABCD$ bisect each other at O)

Thus $\triangle OAP \sim \triangle OCQ$ (by AA similarity), and in fact $\triangle OAP \cong \triangle OCQ$ (by ASA congruence).

Step 2: Deducing $OP = OQ$ [1 Mark]

Using corresponding sides of the similar triangles:

$$\frac{OP}{OQ} = \frac{OA}{OC} = 1 \implies \boxed{\therefore OP = OQ \quad \text{[Hence Proved]}}$$

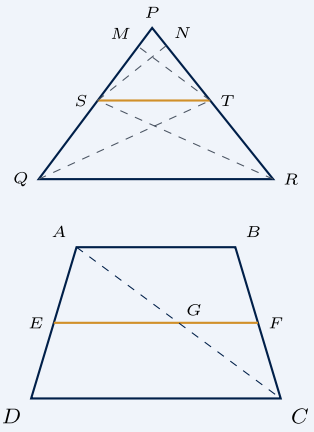
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Q11. State and prove Basic Proportionality Theorem (Thales' Theorem).
Application: In trapezium $ABCD$ with $AB \parallel DC$, a line EF parallel to AB is drawn intersecting AD at E and BC at F. Using Thales' Theorem, prove that $\frac{AE}{ED} = \frac{BF}{FC}$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Part I: Basic Proportionality Theorem [3 Marks]
Statement:

Formula / Theorem Used: If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, the other two sides are divided in the same ratio.



Given: In $\triangle PQR$, line $ST \parallel QR$ meets PQ at S and PR at T.
To Prove: $\frac{PS}{SQ} = \frac{PT}{TR}$.
Construction: Join QT and RS; draw $TM \perp PQ$ and $SN \perp PR$.
Proof: Comparing areas of triangles with the same height:

$$\frac{\text{ar}(\triangle PST)}{\text{ar}(\triangle QST)} = \frac{\frac{1}{2} \cdot PS \cdot TM}{\frac{1}{2} \cdot SQ \cdot TM} = \frac{PS}{SQ} \quad \text{— (i)}$$

$$\frac{\text{ar}(\triangle PST)}{\text{ar}(\triangle RST)} = \frac{\frac{1}{2} \cdot PT \cdot SN}{\frac{1}{2} \cdot TR \cdot SN} = \frac{PT}{TR} \quad \text{— (ii)}$$

Since $\triangle QST$ and $\triangle RST$ stand on the same base ST and lie between the same parallels $ST \parallel QR$, $\text{ar}(\triangle QST) = \text{ar}(\triangle RST)$.
Equating (i) and (ii): $\frac{PS}{SQ} = \frac{PT}{TR}$. **[Proved]**

Part II: Application to Trapezium $ABCD$ [2 Marks]

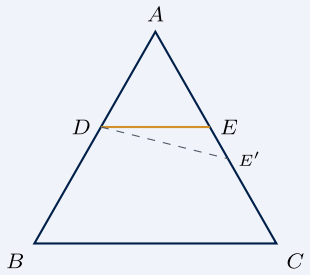
Given: $AB \parallel DC$ and $EF \parallel AB \implies AB \parallel EF \parallel DC$.
Construction: Join diagonal AC intersecting EF at G.
1. In $\triangle ADC$, $EG \parallel DC \implies \frac{AE}{ED} = \frac{AG}{GC}$ (by BPT) — (iii)
2. In $\triangle CBA$, $GF \parallel AB \implies \frac{CG}{GA} = \frac{CF}{FB} \implies \frac{AG}{GC} = \frac{BF}{FC}$ (by BPT) — (iv)
From (iii) and (iv):

$$\therefore \frac{AE}{ED} = \frac{BF}{FC} \quad \text{[Hence Proved]}$$

Q12. State Converse of Basic Proportionality Theorem.

In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $\frac{AD}{DB} = \frac{AE}{EC}$ and $\angle ADE = \angle ACB$. Prove that $\triangle ABC$ is an isosceles triangle. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Step 1: Statement & Proof of Converse of BPT [3 Marks]

Statement:

Formula / Theorem Used: If a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

Suppose line DE divides AB and AC of $\triangle ABC$ such that $\frac{AD}{DB} = \frac{AE}{EC}$. Assume DE is *not* parallel to BC; draw $DE' \parallel BC$ meeting AC at E'.

By BPT on $DE' \parallel BC$: $\frac{AD}{DB} = \frac{AE'}{E'C}$.

Comparing with the given ratio: $\frac{AE}{EC} = \frac{AE'}{E'C}$.

Adding 1 to both sides: $\frac{AE + EC}{EC} = \frac{AE' + E'C}{E'C} \implies \frac{AC}{EC} = \frac{AC}{E'C} \implies EC = E'C$.

Hence E and E' coincide, proving $DE \parallel BC$.

Step 2: Deducing Angle Equality from Converse of BPT [1 Mark]

Here $\frac{AD}{DB} = \frac{AE}{EC}$ is given $\implies DE \parallel BC$ (by Converse of BPT).

Therefore, corresponding angles are equal: $\angle ADE = \angle ABC$ — (i)

Step 3: Proving $\triangle ABC$ is Isosceles [1 Mark]

It is given that $\angle ADE = \angle ACB$ — (ii).

From (i) and (ii): $\angle ABC = \angle ACB$.

Since sides opposite equal angles in $\triangle ABC$ are equal, $AC = AB$.

$\therefore \triangle ABC$ is an isosceles triangle. [Hence Proved]

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Q13. Case Study 07: Mirror Reflection Triangulation of an Ancient Minaret

To determine the height of an ancient minaret, an archaeology team places a flat mirror horizontally on level ground 18 m away from the base of the minaret. A researcher stands 1.5 m behind the mirror such that he can see the reflection of the pinnacle of the minaret in the center of the mirror. The researcher's eye level is 1.8 m above the ground. [4]

(i) **[1 Mark]** Which physical law of optics guarantees that the two triangles formed are equiangular and similar?

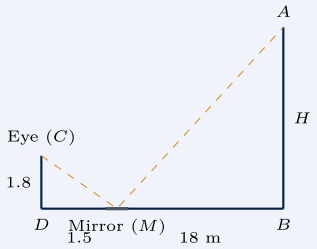
(ii) **[1 Mark]** Name the geometric similarity criterion applicable between the two triangles.

(iii) **[2 Marks]** Calculate the exact height of the ancient minaret above the horizontal ground.

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Let CD be the researcher's eye height (1.8 m), M the center of the horizontal ground mirror, $MD = 1.5$ m, $AB = H$ the minaret height, and $MB = 18$ m.
We know the Law of Reflection and AA similarity property:

Formula / Theorem Used: Angle of incidence = Angle of reflection $\implies \angle AMB = \angle CMD$



Sub-part (i): Physical Law of Optics [1 Mark]

The incident ray from the minaret pinnacle A to mirror M and reflected ray to the eye C obey the **Law of Reflection** (angle of incidence = angle of reflection), making the ground-elevation angles equal ($\angle AMB = \angle CMD$).

Law of Reflection of Light

Sub-part (ii): Similarity Criterion Applicable [1 Mark]

In $\triangle ABM$ and $\triangle CDM$: $\angle ABM = \angle CDM = 90°$ (vertical minaret and observer) and $\angle AMB = \angle CMD$ (by Law of Reflection).

$\triangle ABM \sim \triangle CDM$ by the **AA Similarity Criterion**.

Sub-part (iii): Exact Height of the Ancient Minaret ($AB = H$) [2 Marks]

Since $\triangle ABM \sim \triangle CDM$, corresponding sides are proportional:

$$\frac{AB}{CD} = \frac{BM}{DM} \implies \frac{H}{1.8} = \frac{18}{1.5}$$

Simplifying $\frac{18}{1.5} = \frac{180}{15} = 12$:

$$\frac{H}{1.8} = 12 \implies H = 12 \times 1.8 = 21.6 \text{ m} \implies \boxed{\text{Height of Minaret } = \mathbf{21.6} \text{ m}}$$

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